

Towards intelligent and inclusive mobility for individuals with Autism Spectrum Disorder

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Abstract—This article explores various approaches to support mobility for individuals with Autism Spectrum Disorder, including digital navigation tools, adapted vehicles, immersive technologies, and adaptative tools. We examine the needs these solutions aim to address and their limitations in real-world contexts.

Index Terms—Autism spectrum disorder (ASD), Smart mobility, Digital tools, Mixed reality (MR), Artificial intelligence (AI)

I. INTRODUCTION

Independent mobility is a key factor for social participation and autonomy, yet it remains a major challenge for individuals with Autism Spectrum Disorder (ASD). Sensory sensitivities, difficulties with unpredictability, social interactions, and complex environments can significantly limit access to both public transportation and independent travel. Recent technological advances aimed at supporting autistic individuals in their mobility. This paper provides an overview of existing approaches and discusses their contributions and limitations in real-world contexts.

II. DIGITAL APPLICATIONS FOR NAVIGATION

Digital navigation and journey-planning applications are widely explored as technological supports to enhance independent mobility for individuals on the autism spectrum. These tools, often inspired by mainstream navigation systems, aim to provide clearer, more structured, and simplified guidance adapted to the cognitive and sensory specificities associated with ASD. Core functionalities typically include step-by-step navigation, time management through alerts, and the anticipation of key elements along the route.

Several applications also incorporate real-time adaptation capabilities, allowing dynamic route adjustments in response to traffic conditions or service disruptions. By integrating traffic evaluation, they may reduce exposure to crowded environments, while autism-oriented design features—such as pictograms, simplified instructions, and reassurance mechanisms—seek to limit stress and support comprehension. Positive reinforcement strategies are sometimes used to encourage repeated use and confidence in independent travel.

Empirical studies suggest that such applications can improve journey understanding and reduce travel-related anxiety by increasing predictability and perceived control [1]. These results highlight the potential contribution of digital tools to broader objectives of autonomy and inclusion in mobility.

Nevertheless, important technical limitations remain. Insufficient spatial precision and incomplete or outdated data can affect the reliability of guidance, particularly in dynamic or disrupted transport contexts, which may in turn increase uncertainty and stress for users [1].

Beyond these issues, the literature reveals a gap between application functionalities and the needs expressed by individuals with ASD. User-reported priorities largely concern environmental, social, and organizational aspects of public transport—such as crowd management, rule enforcement, staff support, and system-level accessibility—rather than navigation assistance alone. These needs extend beyond what application-based solutions can address [2].

Overall, while digital applications can support route comprehension and individual coping strategies, they remain insufficient to overcome many structural barriers to public transport use. This limitation points to the relevance of complementary approaches embedded within transport systems themselves, which are examined in the next section.

III. ADAPTED VEHICLES AND ONBOARD ASSISTANCE

Adapted vehicles are often considered as a means for people with ASD to gain greater autonomy in their mobility. However, although many people with ASD express interest in driving (e.g., about two-thirds of young adults), and nearly a third already hold a driving licence in the US, driving a car remains a significant challenge. Environmental stimuli, coupled with having to focus on the same task for a long time, and the need to take into account multiple parameters, can easily lead to sensory overload. The fear of an accident alone discourages a lot of autistic individuals from attempting independent travel.

Many individuals rely on a caregiver to drive them safely. Autism-centered carpooling has also been considered. However, the caregiver cannot fully attend to their passenger's

needs. Moreover, such arrangements do not provide the level of autonomy sought by individuals with ASD.

As such, a study conducted by the University of Michigan [3] revisited the potential of autonomous vehicles. The authors suggest that a driverless car, combined with a soothing in-vehicle application, as well as an interior designed to be stress-free (e.g., removing the divider, as it might create negative emotions), can support greater autonomy while enhancing passenger comfort.

Unfortunately, several challenges come along with this approach. Developing safe and reliable driverless vehicles is not only a technological hurdle but also restricted in many countries due to legal constraints. Furthermore, such solutions may be largely inaccessible to individuals without financial support. These issues highlight the need for further research on vehicle adaptations specifically for individuals with ASD, a domain that remains largely underexplored.

IV. MIXED REALITY AND IMMERSIVE TECHNOLOGIES

Mixed reality (MR), combining augmented and virtual reality, offers significant potential to support the mobility of people with ASD. By overlaying digital information onto the real environment, it helps reduce uncertainty, structure spatial transitions, and limit sensory overload—difficulties frequently reported during travel [4].

Solutions such as Hol’Autisme, which uses Microsoft HoloLens glasses, demonstrate how MR combined with artificial intelligence can analyze and train sensory-motor and social skills [4]. The proposed modules (e.g., PopBalloons, Smile) help develop coordination, sensory stability, and facial expression recognition—skills indirectly related to independent mobility. However, these tools remain largely confined to supervised therapeutic settings and are rarely evaluated in real transport situations.

The Empowered Brain program, based on Google Glass-type devices, offers functionalities more directly related to travel. Modules such as Transition Master allow users to explore unfamiliar places in advance to reduce anxiety related to transitions, while Paint My World aims to filter the visual environment to limit sensory overload [5]. These approaches illustrate the value of MR in preparing for and regulating mobility experiences, despite technical and social limitations (autonomy, device acceptability).

In addition, guidance systems originally designed for visual impairments—such as Wayfindr, NavCog, NaviLens, or Microsoft Soundscape—show strong potential for adaptation to the needs of people with ASD [6], [7]. They facilitate indoor orientation, management of transfers, and access to contextual information, helping to secure travel [8]. Tools like XRAI Glass, which displays audio announcements in real time as text, also address auditory comprehension difficulties that are common in transport environments [9].

Despite these advances, the use of MR for the daily mobility of adults with ASD remains limited. The main barriers include the cost of devices, cognitive fatigue, a lack of validation in real-world conditions, and ethical issues related to data use

[10]. Nevertheless, the future integration of MR with mobile applications and intelligent transport systems represents a promising pathway toward truly inclusive mobility.

V. ADAPTIVE TOOLS AND ASSESSMENT FRAMEWORKS

People with Autism Spectrum Disorder experience difficulties that may affect communication, social interactions, or sensory processing. However, symptoms can vary considerably from one individual to another. As a result, adaptive tools represent a promising approach, it makes it possible to design services that respond to the specific needs of each autistic person in order to promote their autonomy, comfort, and inclusion. These tools are therefore mainly analytical tools that help identify the characteristics of an individual’s autism.

For example, the TSAT (Transportation Skills Assessment Tool) is a tool, which is still under development. It consists of a test made up of 37 tasks aimed at observing a person’s cognitive and practical abilities in transportation-related situations. Although the tool appears to be functional, the pilot study involved only a very small number of participants. Therefore, the results are preliminary and cannot be generalized to all adults with autism. Moreover, the TSAT does not create transportation services, it merely assesses access to existing ones. If an adapted transportation network does not exist in the geographical area, the tool remains of limited interest. [11]

Another interesting tool is Mind the Gap. This tool aims to assess the accessibility of public transportation for autistic people. The objective is to identify sensory, organizational, or social barriers in order to propose better-adapted spaces, services, and products. However, this tool remains a thesis and is not (yet) a widely operational or generalized service. The tool exists, but it has not necessarily been implemented within transportation networks. [12]

Thus, TSAT and Mind the Gap are solutions that lack experience in “real-life” situations. Moreover, taken individually, these tools are not sufficient to transform the mobility experience of autistic people. However, when combined, they offer a more comprehensive perspective: one evaluates individual skills, while the other analyzes the environment. Together, they pave the way for truly inclusive services.

VI. CONCLUSION AND PERSPECTIVE

This paper reviewed existing approaches to support the mobility of individuals with Autism Spectrum Disorder, including digital navigation applications, adapted vehicles, mixed reality technologies, and adaptive assessment tools. These solutions aim to reduce uncertainty, sensory overload, and anxiety, while fostering greater autonomy in daily travel.

The investigation shows that while digital and immersive tools can improve route comprehension and preparation, they remain limited by technical constraints. Similarly, autonomous vehicles present promising opportunities but face significant technological, legal, and economic challenges. Furthermore, these solutions focus mostly on individual-level adaptation.

Overall, applications and mixed reality seem to be the most promising approaches towards inclusive mobility.

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